

Module 10

ML: Intro to machine learning

Intro to classical ML

- ~~Review~~ Already Learned - Revision
- To learn - Overview

1. What is ML?

2. ML vs SDE

3. ML tasks

4. Type of learning

• Application

• EDA, Feature Engineering

↓

exploratory data analysis

Revision

Let's briefly discuss what we learnt in previous modules

1. Data library

learnt how to load & process the data

e.g. numpy, pandas, matplotlib, seaborn

2. probability & statistics

learnt how to understand data

3. co-ordinate geometry & linear algebra

learnt how to work with high dim spaces

4. Calculus & optimization

optimisation technique like GD,

Programming

↳ follow rules.

e.g. sorting

input :- [1 7 3 4]

output :- [1, 3, 4, 7]

here rules will be

for (i=0, length)

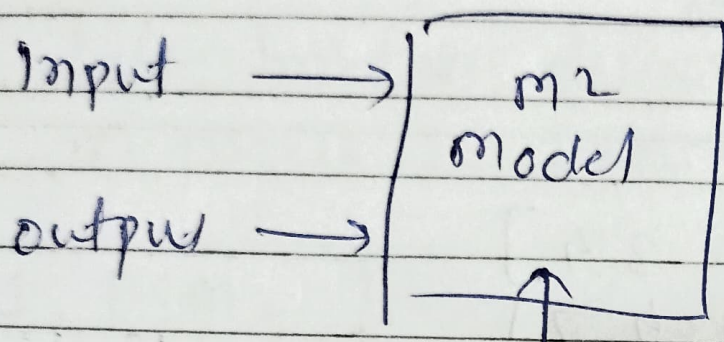
if (input[i] < min)

swap

⋮

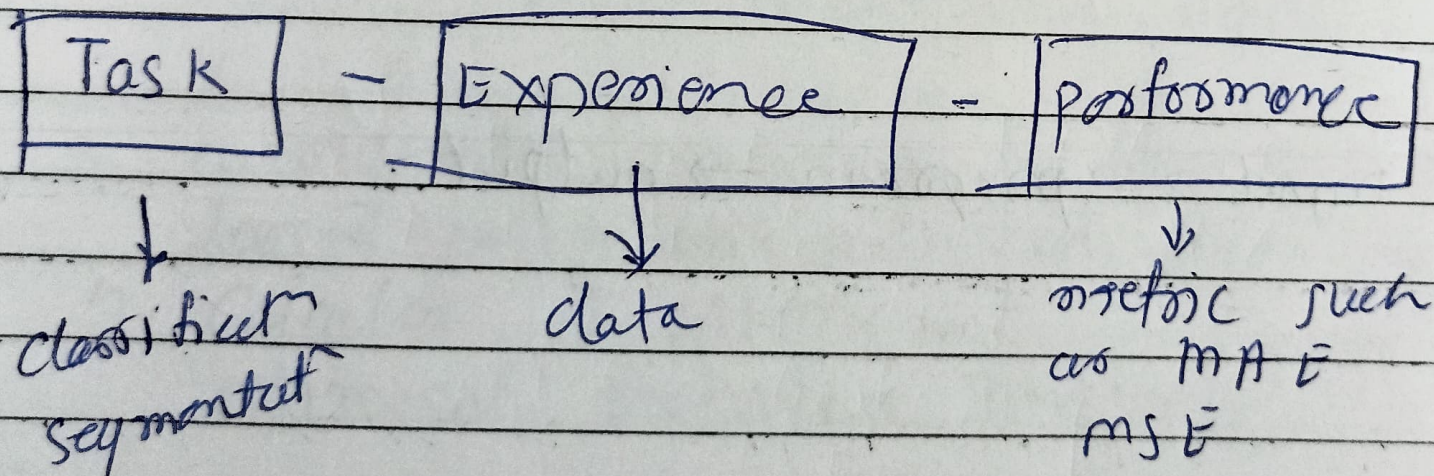
input → program → output

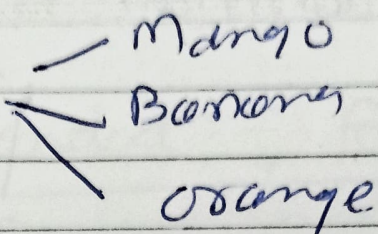
ML : -



learn to get output from input

Tom Mitchell's Defn



① task classifity 
Mango
Bananas
orange

② Data pictures + labels

③ performance top5 top2

study algorithm which

improve performance (p/

A task (T)

with experience (E)

ML classifier

classifier

fit
unit

Regression

predict house
price

↓ finite o/p value
always have

↓ infinite o/p value
may have

This is learned by supervised learning

Clustering

~~We have~~ data but no label, &
we want to divide these data
based on some features.

This is called clustering

This is ~~also~~ ~~called~~ learned by Unsupervised
learning

Another case of ml

Recommendation, forecasting → time bound

